MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES 5U4G AND 5Z3 1/ 2/
a b

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Rectifier, full-wave, high vacuum

Outline	---	a--16-3 (EIA) b--16-1 (EIA) Ⓕ
Base	---	a--B5-15 b--A4-9
Envelope	---	a, b--ST-16
Cathode	---	a, b--Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	a nc	f	---	2p	---	1p	---	f
		b f	2p	1p	f	---	---	---	---

ABSOLUTE-MAXIMUM RATINGS: See note 1

Parameter:	Ef	Epp/p	epx	ib	Io	RL	CL	i(surge)	Ⓕ Alt
Unit:	Vac	Vac	v	ma	mAdc	Ohms	μF	a	ft
Maximum:	5.5	(see note 2)	1,540	880	(see note 3)	---	---	4.4	10,000
Minimum:	4.5	---	---	---	---	---	---	---	---
TEST CONDITIONS:	5.0	500	---	---	---	2,100	8	---	---

GENERAL:

Qualification - Required

1/ See note 7

2/ To identify those tests that are applicable to a given type or to several types; tube types are designated by letters.

Ⓕ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
1231	Emission (1)	E2b = 0; E1b = 60 Vdc	0.65	II	I1s	260	420	mAdc
1231	Emission (2)	E1b = 0; E2b = 60 Vdc	0.65	II	I2s	260	420	mAdc
1353	Operation of rectifiers	See note 4	0.65	II	Ic	230	---	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1301	Filament current		6.5	S3	If	2.7	3.3	A
---	Alinement	See note 5	6.5	S3	---	---	---	---
1101	Secureness of base, base insert, or cap		6.5	S3	---	---	---	---
1111	Base pin solder depth	See note 8	6.5	S3	---	---	---	---
1211	Insulation of electrodes		4.0	S3	R	10	---	Meg
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life (500 hours)	Group B (see note 6)	---	---	---	---	---	---
---	Life-test end point (500 hours)	Operation of rectifiers	---	---	Io	202	---	mAdc

NOTES:

1. Tubes are to be operated in a vertical position. Tubes may be mounted in a horizontal position provided pins No. 1 and 4 for type a, and pins No. 1 and 2 for type b are in a vertical plane.

2. All values of Epp/p refer to the unloaded supply voltages and the rectifier operation with sinusoidal supply voltages, within the frequency range of 25 to 1,000 hertz.

3. To simplify the application of the maximum ratings to circuit design, the absolute-maximum ratings are presented in chart form as rating charts I, II, and III. Rating chart I presents the absolute-maximum ratings for ac anode-supply voltage and dc output current. Rating chart II provides a convenient method for checking conformance with the absolute-maximum steady-state peak-anode current rating. Rating chart III offers a convenient method for checking conformance with the absolute-maximum transient peak-anode current rating.

   With a capacitor-input filter, the conditions of each rating chart must be satisfied in order to obtain performance within all of the electrical absolute-maximum ratings. With a choke-input filter, operation within the indicated boundary of rating chart I will assure performance within all of the appropriate electrical absolute-maximum ratings.

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NOTES: -Continued

3. -Continued

(a) Rating chart I (see figure 1). To determine whether a given set of operating conditions is within the maximum ratings presented in rating chart I, the operating point of ac supply voltage (rms) per anode and dc output current per anode for the application should be entered on the chart. If, for choke-input-filter operation, this point falls within the boundary FABCDG, operation within the ratings of ac supply voltage and dc output current is assured. For capacitor-input-filter operation, the operating point must fall within the boundary FAEDG.

If the operating point falls outside of the prescribed area of permissible operation, either the dc output current or the ac supply voltage (rms) must be reduced until operation within the area of permissible operation is obtained. The type of corrective action to be taken necessarily depends on the requirements of the application.

(b) Rating chart II (see figure 2). To determine whether a given set of operating conditions is within the maximum ratings presented in rating chart II, the operating point of dc output current per anode and rectification efficiency should be entered on the chart. If this point falls within the shaded area of the chart, operation within the absolute-maximum steady-state peak-anode current rating is assured.

If the operating point falls outside of the prescribed area of permissible operation, corrective action must be taken. For a given ac supply voltage and dc output current, the rectification efficiency can usually be reduced either by increasing the total supply resistance per anode or by decreasing the filter input capacitor.

(c) Rating chart III (see figure 3). To determine whether a given set of operating conditions is within the maximum ratings presented in rating chart III, the total Rs is first calculated from measurements of Rpri, Rsec, RA, and N. The operating point of Rs and ac supply voltage is then entered on rating chart III. If the total anode supply resistance is equal to or greater than the maximum value of Rs, as indicated by the curve, operation within the maximum transient peak-anode current rating is assured. If the conditions of rating chart III are not satisfied, additional resistance should be added in series with each diode anode so that the indicated minimum value of Rs is obtained.

It should be noted that the anode supply resistance serves to limit both the steady-state anode current and the transient peak current. The minimum value of resistance required for these functions will not necessarily be the same. The larger value of resistance must be used.

Series inductance in the anode supply is also effective in limiting the transient peak-anode current. If appreciable series inductance is present in the anode supply, a smaller-than-indicated value of series-anode supply resistance may be used providing it is determined experimentally that the combined effect of the available supply resistance and inductance is adequate to limit the transient peak-anode current to its rated maximum value.

4. Test in a full-wave circuit, with total source resistance, including transformer and series resistance adjusted to 100 ohms. To measure transformer source resistance (see figure 4) proceed as follows:

(a) Remove circuit from power source.

(b) Short the circuit at the point where power source is normally applied.

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NOTES: -Continued

4. -Continued

(c) Apply sufficient 60 Hz ac voltage to one-half of the secondary of the transformer through a milliammeter such that the measured current is 0.1 ampere. CAUTION: If transformer is not capable of 0.1 ampere, use some other convenient lower current.

(d) Measure the ac volts across the transformer half-secondary using a vacuum tube voltmeter of not less than 100,000 ohms input impedance.

(e) Calculate the resistance from the expression

$$R\ (\text{ohms source}) = \frac{E\ (\text{volts})}{I\ (\text{amperes})}$$

Add resistance in series with transformer common center tap such that the sum of the transformer source resistance and the added resistance is 100 ohms.

5. The plane of the filament shall be parallel, within ±22.5°, to the plane passing through the centerline of pins No. 1 and 4 for type a, and pins No. 1 and 2 for type b.

6. In life-test operation the values of RL and CL given in test conditions may be considered as approximate, and shall be adjusted on a bogie tube to give Ic = 250 mAdc with ib equal to or greater than 800 ma.

7. Tube type 5X4G is hereby deleted from this tube specification sheet. Tubes for replacement purposes will be procured to EIA data, Release Number 73, Type 5X4G.

8. This test applies for flexible leads as well as for rigid leads.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

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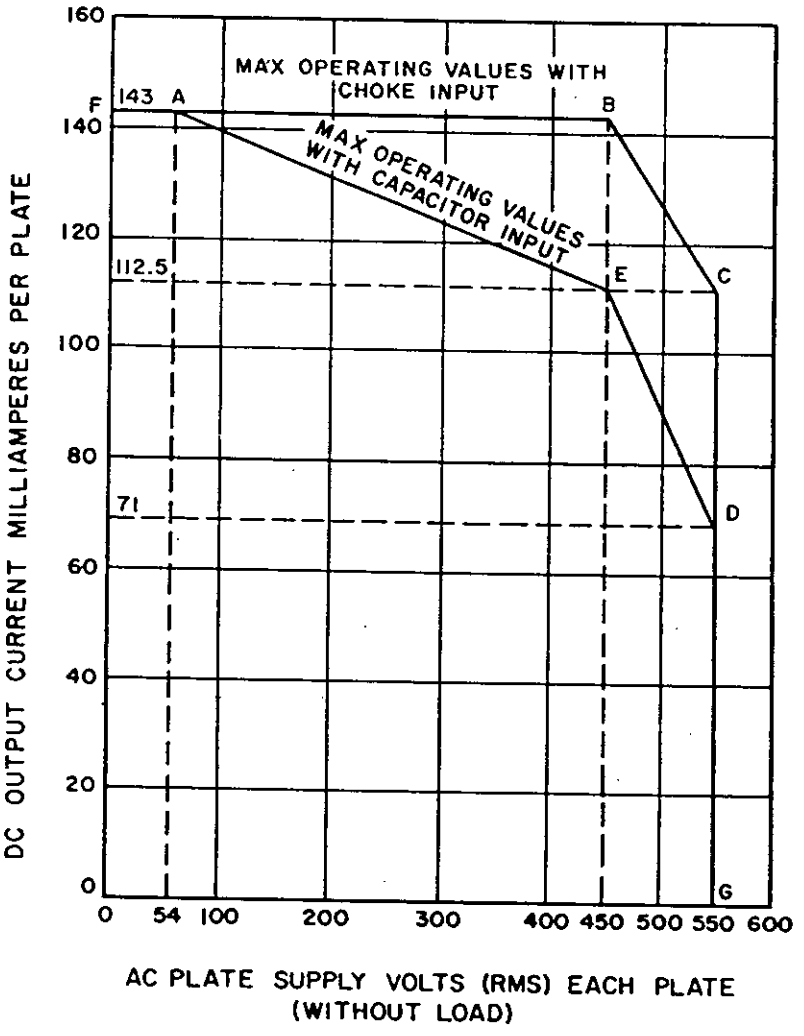


FIGURE 1. Rating chart I.

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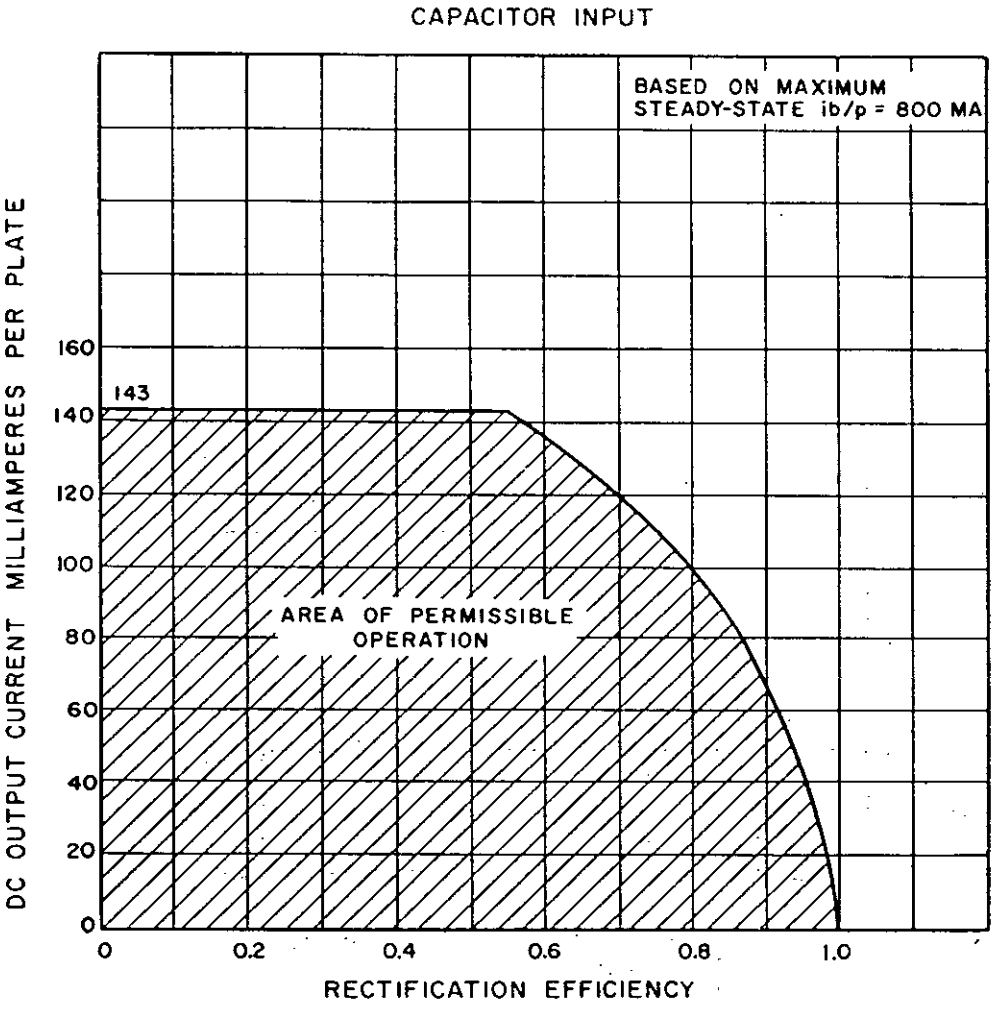


FIGURE 2. Rating chart II.

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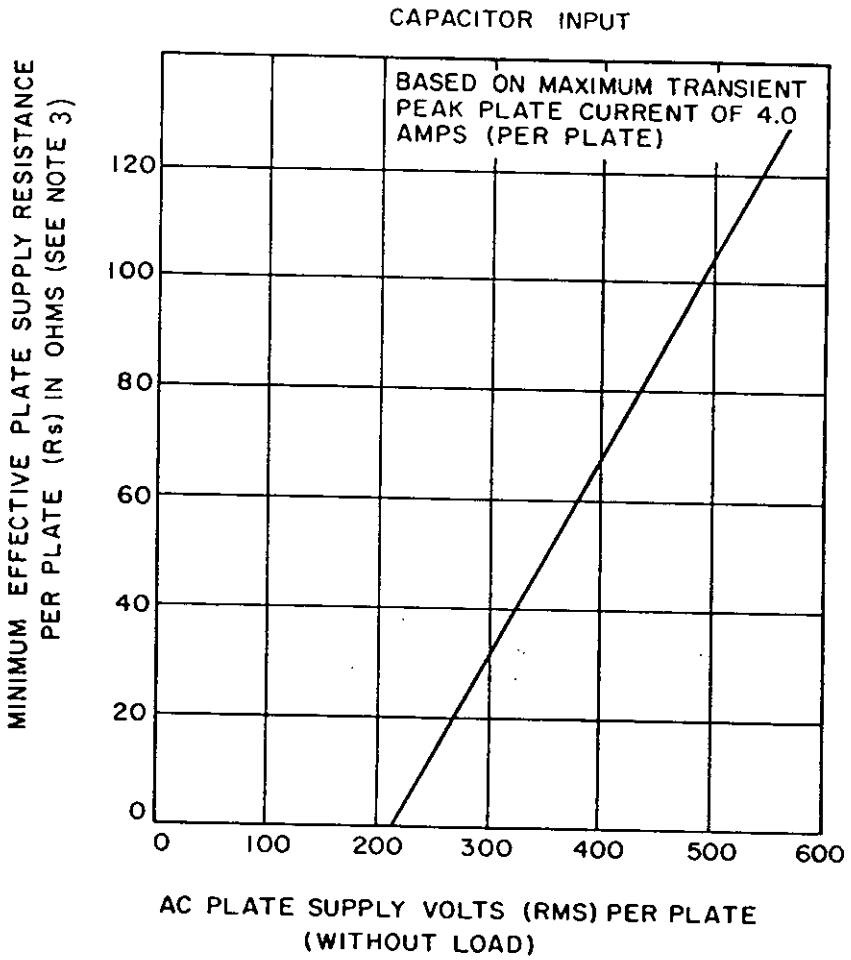


FIGURE 3. Rating chart III.

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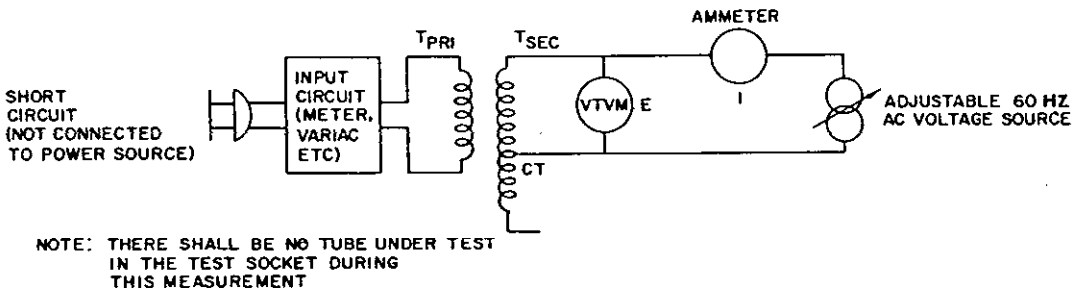


FIGURE 4. Schematic diagram for measuring source resistance.

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